RAKHIMBOEV BOBUR

SOFTWARE & AI ENGINEER

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Software & AI Engineer building LLM systems that run inside real engineering software. At BSP Engineering I connect AI agents to Autodesk Revit through an MCP server and automate fabrication isometrics, weld lists and BOMs for semiconductor-fab piping. Before that, four years of full-stack and DevOps work with NestJS, Next.js and Docker. Independent work covers cross-lingual RAG, n8n retrieval pipelines and a real-time multiplayer game.

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WORK EXPERIENCE

Software & AI Engineer — BSP Engineering | Pyeongtaek, Korea — June 2026 – Present

- Built BSP.ISO, a pyRevit extension that turns Revit piping models into fabrication-ready isometric drawings for semiconductor-fab utility (PCW) piping: automatic weld numbering, spool splitting, BOM and weld-list tables, one-click DWG export.
- Built a Python MCP server that connects Revit to AI agents, so models can be inspected and edited through plain-language instructions.
- Wrote rule-based weld classification (shop, field and random-length welds) and spool logic that works on every line in a model, not on hard-coded cases.
- Built Room Box, a modular Revit plugin that calculates room volumes, AC unit counts and other MEP quantities.
- Evaluated Autodesk APS Design Automation as a path to running the tools as a cloud service.

FullStack & DevOps Engineer — Mimsoft | Tashkent, Uzbekistan — March 2023 – March 2025

- Built and maintained customer-facing platforms and internal business systems with NestJS, Node.js, React and Next.js.
- Integrated OpenAI and other LLM APIs into production backends.
- Designed MongoDB schemas and implemented authentication with JWT, bcrypt and session-based security.
- Ran Docker-based deployments, managed production environments and fixed performance problems that only appeared under real traffic.

Backend & Frontend Developer — Boldare | Poland — January 2021 – January 2023

- Built web applications with NestJS, Node.js, React and Next.js for enterprise and client-facing platforms.
- Moved deployments to Docker-based automation, cutting deployment time by about 40%.
- Migrated backend services from REST to GraphQL, improving API efficiency and frontend data handling by 20–40%.
- Built reusable frontend components and improved application responsiveness.

SELECTED PROJECTS

- Multilingual RAG (github.com/Max19942008/RAG-forr-Multi-Language-): upload a document in one language, ask in another, and get an answer that cites the exact passage and page. Chunking is decided per script (Latin, Cyrillic, Korean, CJK) and every chunk is measured with the real tokenizer, so no text is silently cut. Cross-lingual retrieval with multilingual-e5-large.
- SurveyFlow (github.com/Max19942008/surveyflow): survey platform for master's and PhD students with an AI assistant built as an n8n RAG pipeline: AI-drafted questionnaires and a methodology chat that cites its sources. Next.js 14, NestJS 10, MongoDB, Qdrant, Ollama (bge-m3), Gemini, Docker Compose, nginx.
- Global Challenge Chess (github.com/Max19942008/global-challenge-chess): cross-platform online chess. Flutter app for Android and iOS in three languages with a Stockfish bot; real-time game server with matchmaking, server-side clock, Glicko-2 rating and reconnection (Redis, PostgreSQL); shared TypeScript rules package for server and web.

TECHNICAL SKILLS

- AI / LLM: OpenAI, Claude and Gemini APIs, RAG, AI agents, MCP (Model Context Protocol), n8n, Qdrant, Ollama, Hugging Face, Sentence-Transformers, PEFT / LoRA, PyTorch, OpenCV
- Engineering software: Revit API, pyRevit, C#, IronPython, Autodesk APS
- Backend: Python, FastAPI, Node.js, NestJS, Express, REST, GraphQL, WebSocket
- Frontend & mobile: TypeScript, React, Next.js, Flutter, React Native
- Data: PostgreSQL, MongoDB, Redis, MySQL
- DevOps: Docker, Docker Compose, Nginx, Linux, PM2
- Languages: Uzbek (native), English, Korean (TOPIK Level 5), Russian, Turkish

EDUCATION

- PhD (in progress) — Joongbu University, South Korea | Sep 2024 – Present
- Master's degree — Woosong University, South Korea | 2019 – 2021
- Bachelor's degree — Tashkent Engineering University, Uzbekistan | 2014 – 2018